What Changes When LLMs Learn to Reason? Mechanistic Signatures of Cognitive Bias Processing

Anonymous Authors

Abstract

Does reasoning training change how LLMs internally represent cognitive biases, or merely suppress biased outputs? We investigate using matched-architecture model pairs that share identical structure but differ in reasoning training: Llama-3.1-8B-Instruct vs. R1-Distill-Llama-8B, and OLMo-3-7B-Instruct vs. OLMo-3-7B-Think. Behaviorally, reasoning training dramatically reduces heuristic lure rates across architectures: 27.3% to 2.4% (Llama) and 14.9% to 0.9% (OLMo). We probe residual stream activations with linear classifiers to separate conflict from no-conflict items across 235 matched pairs spanning 11 cognitive bias categories. The standard Llama model achieves near-perfect conflict/control separability (peak AUC = 0.974, 95% CI [0.952, 0.992]), but the reasoning model shows *lower* separability (AUC = 0.930 [0.894, 0.960]; non-overlapping CIs)—the internal heuristic/deliberative boundary is *blurred*, not sharpened, by reasoning training. Cross-prediction confirms specificity: a probe trained on vulnerable categories yields transfer AUC = 0.378 on immune categories, and cross-model probe transfer (Llama→R1: 0.920; R1→Llama: 0.954) shows the processing-mode direction is shared across reasoning-differentiated models. First-token confidence analysis confirms De Neys' conflict-detection prediction: reduced probability and elevated entropy on conflict items. Attention analysis reveals the reasoning model has nearly twice as many S2-like-specialized heads (57 vs. 30; BH-FDR corrected), suggesting redistribution of deliberative computation across the network. A within-model dissociation using Qwen 3-8B's thinking toggle sharpens the picture: inference-time thinking reduces lure rates (21% to 7.3%) without changing pre-generation probe signatures (both AUC = 0.971). Training changes S1-like/S2-like encoding; inference-time thinking does not. Finally, natural frequency framing—which reduces base rate neglect in humans—paradoxically *increases* it in both models, demonstrating that ecological rationality does not transfer to LLMs.

1 Introduction

Kahneman's dual-process framework distinguishes fast, heuristic processing ("System 1") from slow, deliberative processing ("System 2") (**?**). While the binary framing is an acknowledged oversimplification—modern accounts treat this as a *continuum* of processing intensity (**???**)—the core observation that cognitive systems flexibly allocate effort, with measurable consequences for bias susceptibility, remains robust. Crucially, De Neys' conflict detection paradigm (**??**) demonstrates that humans often *detect* conflicts between heuristic and normative responses even when they fail to override the heuristic—a dissociation between conflict *monitoring* and conflict *resolution* with clear neural correlates in the anterior cingulate cortex (**?**).

LLMs exhibit analogous patterns: instruction-tuned models reproduce human-like errors on conflict items while succeeding on matched controls (**?**), and reasoning-trained models resist many such lures (**?**), though the protection is bias-specific rather than universal (**?**). A growing literature documents these parallels (**???**), and **?** showed that linear probes can detect and steer cognitive biases via internal representations.

Yet behavior alone cannot distinguish genuine processing-mode differences from surface mimicry—chain-of-thought traces are often unfaithful, with only ∼2.3% of reasoning steps causally influencing the final answer (**???**). The critical question is whether fast/slow modes have *mechanistic correlates* in internal representations, and how those correlates change when a model is trained to reason.

We address this question with a natural experiment: Llama-3.1-8B-Instruct and DeepSeek-R1-Distill-Llama-8B share identical architectures (32 layers, 32 attention heads, 4096 hidden dimensions) but differ in reasoning distillation training. Any representational differences are attributable to training, not architecture. We make five contributions:

1. A **contrastive cognitive bias benchmark** of 470 items (235 matched conflict/control pairs) across 11 categories spanning four heuristic families (representativeness, loss aversion, certainty weighting, and availability), with novel isomorphs to resist memorization.
2. **Behavioral validation** showing a dramatic gap: 27.3% overall lure rate (standard) vs. 2.4% (reasoning), replicated on OLMo-3-7B (14.9% $\rightarrow$ 0.9%), with category-level specificity revealing which biases are vulnerable and model-specific vulnerability profiles (e.g., loss aversion in OLMo but not Llama). First-token confidence analysis confirms De Neys' conflict-detection prediction: reduced probability and elevated entropy on conflict items.
3. **Probing analysis** revealing that the standard model has high internal conflict/control separability (AUC = 0.974, 95% CI [0.952, 0.992]) while the reasoning model *blurs* this distinction (AUC = 0.930, 95% CI [0.894, 0.960]; non-overlapping CIs), with cross-prediction, transfer matrix, and cross-model probe transfer (Llama→R1: 0.920; R1→Llama: 0.954) confirming specificity to a shared processing-mode direction.
4. **A dissociation between training and inference**: Qwen 3-8B's thinking toggle changes behavior (21% $\rightarrow$ 7.3% lure rate) but produces *identical* probe signatures (AUC = 0.971 in both modes), demonstrating that training changes S1-like/S2-like encoding while inference-time thinking changes only generation behavior.
5. **Sparse autoencoder features surviving falsification**: 41 SAE features show significant differential activation between conflict and control items (BH-FDR $q < 0.05$), with all 41 surviving Ma et al. falsification testing (**?**)—zero spurious features—grounding the probe-level findings in interpretable feature directions.

Our framing: **Same Architecture, Different Minds.** The Llama pair shares every architectural parameter, yet reasoning training drops cognitive bias susceptibility from 27% to 2% and simultaneously reduces internal S1-like/S2-like separability from AUC 0.974 to 0.930 (non-overlapping 95% CIs). The OLMo-3-7B family independently replicates this pattern (14.9% $\rightarrow$ 0.9%), confirming generality across architectures. Qwen 3-8B reveals the converse: thinking mode changes behavior (21% $\rightarrow$ 7.3%) without changing the pre-generation representation (AUC = 0.971 in both modes). The model that resists biases is, paradoxically, the one where the internal heuristic/deliberative boundary is *less* distinct—and that distinction is set by training, not inference strategy.

2 Benchmark

We construct a contrastive cognitive bias benchmark following three design principles: (1) every conflict item has a structurally matched no-conflict control (**?**); (2) all items use novel surface features to resist memorization; and (3) category diversity enables specificity analysis.

The benchmark comprises **470 items** organized as **235 matched pairs** across 11 categories spanning four heuristic families:

- **Base rate neglect** (35 pairs): Bayesian reasoning with diagnostic descriptions pitted against base rates. Includes 10 pairs in Gigerenzer's natural frequency format ("10 out of 1000" instead of "1%") to test robustness to the ecological rationality critique (**?**).
- **Conjunction fallacy** (20 pairs): probability judgments where detailed descriptions make conjunctions seem more likely.
- **Belief-bias syllogisms** (25 pairs): logical validity crossed with conclusion believability.
- **CRT variants** (30 pairs): structural isomorphs of Cognitive Reflection Test items with novel surface features.
- **Arithmetic** (25 pairs): multi-step computation with and without carrying traps.
- **Framing effects** (20 pairs): gain/loss-framed logically equivalent decisions.
- **Anchoring** (20 pairs): numerical estimation with extreme vs. neutral anchors.
- **Sunk cost fallacy** (15 pairs): decision scenarios where large prior investment tempts continuation despite negative expected value (loss aversion family).

- **Loss aversion** (15 pairs): risky gambles where loss-framed and gain-framed options have identical expected value.
- **Certainty effect** (15 pairs): preference for certain outcomes over probabilistically equivalent or superior gambles (certainty weighting family; Kahneman & Tversky, 1979).
- **Availability heuristic** (15 pairs): frequency and risk judgments where vivid or recent exemplars bias estimation (availability family).

Of the final four categories, sunk cost (0% lure, both models), certainty effect (0%, both), and loss aversion (0% Llama; 33% OLMo Instruct, 0% OLMo Think) are validated. Availability shows a Llama-specific vulnerability (13.3% lure) but 0% for R1-Distill. The 10 natural frequency base rate pairs have been validated (see Discussion). All primary analyses use the original 7-category, 165-pair subset (330 items) plus the OLMo cross-architecture replication.
Each item is scored on a four-way classification: *correct*, *S1-lure* (the specific heuristic-predicted wrong answer), *other-wrong*, or *refusal*. Classic problems (e.g., the original bat-and-ball) are included only as contamination baselines and excluded from all primary analyses.

3 Methods

3.1 Models

Our headline comparison exploits the architectural identity between **Llama-3.1-8B-Instruct** (**?**) and **R1-Distill-Llama-8B** (**?**): both have 32 transformer layers, 32 query heads (8 KV heads via grouped-query attention), and 4096 hidden dimensions. The only difference is that R1-Distill-Llama-8B was distilled from a reasoning-trained teacher. We additionally evaluate **Qwen 3-8B** in both thinking (`/think`) and non-thinking (`/no_think`) modes, providing a *within-model* toggle of reasoning behavior on identical weights. For cross-architecture replication, we evaluate **OLMo-3-7B-Instruct** and **OLMo-3-7B-Think** (**?**), which share the same OLMo base but differ in reasoning training—analogous to the Llama pair but from an independent architecture family.

3.2 Activation Extraction

For each model–item pair, we extract residual stream activations (post-MLP, post-residual) at all 32 layers at position `P0`: the last prompt token before generation. This "pre-decision" position captures the internal state from which the answer is generated. Activations are stored in BFloat16 (HDF5).

3.3 Linear Probes with Hewitt–Liang Controls

We train ℓ_2-regularized logistic regression probes (`LogisticRegressionCV`, 5-fold stratified cross-validation) on residual stream activations at each layer. The probe target is binary: *conflict* vs. *no-conflict* condition, restricted to the three vulnerable categories (base rate neglect, conjunction fallacy, syllogisms) where the standard model shows non-trivial lure rates.

Controls. We apply Hewitt–Liang control tasks (**?**): probes trained on randomly permuted labels establish a selectivity floor. Selectivity (true AUC minus control AUC) below 5 percentage points indicates the signal reflects probe capacity, not representational structure. We report ROC-AUC as the primary metric with bootstrap 95% confidence intervals (1000 resamples).

Immune categories. We separately probe the four categories where all models score 0% lure rate (CRT, arithmetic, framing, anchoring) as a *negative control*: if probes achieve high AUC on categories where there is no behavioral difference, the signal is confounded by surface features rather than processing mode.

Table 1: S1-lure rates (%) on conflict items by model and category. Bolded numbers indicate vulnerable categories (lure rate >10%). OLMo columns provide cross-architecture replication from an independent model family. Loss aversion is vulnerable in OLMo Instruct (33%) but immune in Llama (0%); availability shows a Llama-specific vulnerability (13%) absent in all other models. Natural frequency base rate framing increases lure rates (Llama: 84% → 100%; R1-Distill: 4% → 40%). — = not yet evaluated on the expanded set.

	Llama	**R1-Distill**	**Qwen 3-8B**		**OLMo**	**OLMo**
Category	**8B-Inst.**	**Llama-8B**	**no think**	**think**	**3-7B-Inst.**	**3-7B-Think**
Base rate	**84**	4	**56**	4	**46**	0
Conjunction	**55**	0	**95**	**55**	**50**	0
Syllogism	**52**	0	0	0	0	4
Loss aversion	0	0	0	0	**33**	0
Sunk cost	0	0	0	0	—	—
Certainty eff.	0	0	0	0	—	—
Availability	**13**	0	0	0	—	—
CRT variants	0	3	0	0	3	3
Arithmetic	0	0	0	0	0	0
Framing	0	0	0	0	0	0
Anchoring	0	**10**	**10**	0	5	0
Overall lure	**27.3**	**2.4**	**21.0**	**7.3**	**14.9**	**0.9**

4 Results

4.1 Behavioral Validation

Table **??** reports lure rates (proportion of S1-lure responses on conflict items) across all models and categories.

Four findings emerge. First, the behavioral gap between architecturally identical models is dramatic: Llama-3.1-8B-Instruct produces heuristic lures at a 27.3% rate; R1-Distill-Llama-8B reduces this to 2.4%. Second, vulnerability is *category-specific*: base rate neglect, conjunction fallacy, and syllogisms are the primary susceptible categories, while CRT, arithmetic, framing, and anchoring are largely immune (with sporadic low-level lure rates ≤10% in some models). Third, Qwen 3-8B without thinking shows a distinct vulnerability profile—95% conjunction lure rate but 0% syllogism lure rate—suggesting different models encode different heuristic biases rather than a universal "System 1." Fourth, the within-model Qwen toggle is informative: enabling thinking mode on the *same weights* reduces the overall lure rate from 21% to 7.3%, with a striking asymmetry across categories. Base rate neglect drops from 56% to 4% (near-complete correction), while conjunction fallacy drops only from 95% to 55% (still majority-susceptible). This dissociation suggests that thinking mode provides a genuine reasoning intervention for some bias types but fails to overcome deeply encoded heuristics for others.

Cross-architecture replication. To test generality, we evaluate OLMo-3-7B-Instruct and OLMo-3-7B-Think (**?**), which share the same base but differ in reasoning training (analogous to the Llama pair). OLMo Instruct reproduces the core vulnerability pattern: 46% base rate and 50% conjunction lure rates, with an overall rate of 14.9%. OLMo Think reduces this to 0.9%—near-complete elimination, even more effective than R1-Distill (2.4%). The pattern replicates across two independent architecture families. Notably, OLMo Instruct reveals a *model-specific* vulnerability: 33.3% lure rate on loss aversion items, where Llama showed 0%. Different training data and procedures expose different biases, consistent with the view that bias susceptibility is not a universal "System 1" property but depends on what heuristics each model has internalized. Probing confirms the mechanistic pattern: OLMo Instruct reaches AUC = 0.996 [0.988, 1.000] at layer 24, while OLMo Think reaches AUC = 0.962 [0.934, 0.982] at layer 22—a statistically significant gap (non-overlapping 95% CIs), the same direction of blurring observed in the Llama pair, with the standard model showing higher separability in 30 of 32 layers.

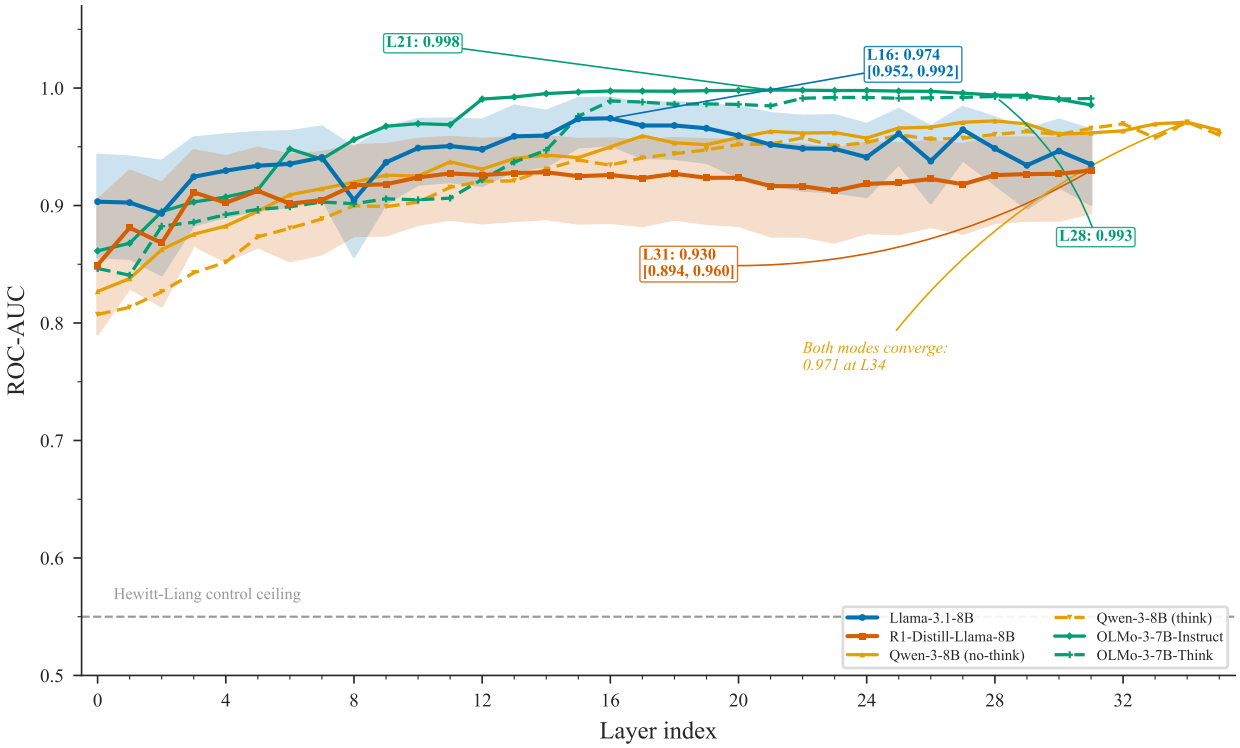

Figure 1: Layer-wise probe ROC-AUC for classifying conflict vs. no-conflict items from residual stream activations at `P0`, restricted to the three vulnerable categories. Llama-3.1-8B-Instruct (blue) peaks at AUC = 0.974 [0.952, 0.992] at L16; R1-Distill-Llama-8B (red) peaks at AUC = 0.930 [0.894, 0.960] at L31—non-overlapping CIs confirm statistical significance. Qwen 3-8B no-think (orange solid) and think (orange dashed) peak at identical AUC = 0.971 at L34—the thinking toggle does not change pre-generation representations. Dashed gray line: Hewitt–Liang control task ceiling.

Implicit conflict detection. The De Neys confidence paradigm (**?**) predicts that conflict detection manifests as reduced confidence even when the heuristic response prevails. Llama's first-token probability is 4.2 percentage points lower on conflict items (0.751 vs. 0.793), and top-10 token entropy is 16% higher (0.948 vs. 0.817). The model exhibits reduced confidence on items where the heuristic cue conflicts with the normative answer, paralleling the human finding that conflict monitoring occurs even when the heuristic response prevails. (R1-Distill always begins generation with `<think>`, making the first-token metric uninformative for reasoning models.)

4.2 Layer-Wise Probe Results

Figure **??** shows layer-wise probe AUC for the conflict/control classification on the three vulnerable categories.

Llama-3.1-8B-Instruct. The conflict/control signal builds rapidly from AUC = 0.85 at layer 0 to AUC = 0.974 at layer 16 (95% CI [0.952, 0.992]), then remains above 0.95 through layer 31. The peak at mid-to-upper layers is consistent with prior findings that mid-network representations encode task-relevant distinctions most strongly (**?**).

R1-Distill-Llama-8B. The layer-wise profile is qualitatively similar—signal builds in early layers—but is *uniformly lower*, peaking at AUC = 0.930 (95% CI [0.894, 0.960]) at layer 31. The CIs do not overlap with Llama's ($p < 0.05$), confirming the gap is statistically significant. The later peak layer (L31 vs. L16) suggests that reasoning distillation shifts conflict-relevant information toward deeper layers.

Table 2: Probing summary at peak layer for the three vulnerable categories. 95% CIs from 1000 bootstrap resamples (Llama and R1-Distill). Selectivity = true AUC − control AUC. Qwen think and no-think produce identical probe results despite a 3× behavioral difference. OLMo replicates the same direction: Instruct > Think.

Model	Peak Layer	Peak AUC	95% CI	Control AUC	Selectivity
Llama-3.1-8B-Instruct	L16	0.974	[0.952, 0.992]	0.63	0.343
R1-Distill-Llama-8B	L31	0.930	[0.894, 0.960]	0.55	0.377
Qwen 3-8B (no think)	L34	0.971	—	—	—
Qwen 3-8B (think)	L34	0.971	—	—	—
OLMo-3-7B-Instruct	L24	0.996	[0.988, 1.000]	—	—
OLMo-3-7B-Think	L22	0.962	[0.934, 0.982]	—	—

Qwen 3-8B (no-think mode). Probes on Qwen 3-8B no-think activations reach peak AUC = 0.971 at layer 34 on the vulnerable categories, intermediate between Llama (0.974) and R1-Distill (0.930) and consistent with its intermediate behavioral vulnerability (21% lure rate).

Qwen 3-8B (think mode). Probes on Qwen 3-8B think-mode activations—extracted at the same `P0` position before the `<think>` block begins—reach *identical* peak AUC = 0.971 at layer 34. Despite a 3× behavioral improvement (lure rate 21% → 7.3%), the pre-generation representation is unchanged—consistent with **?**, who showed probes can predict chain-of-thought success before the first token is generated. This is the key dissociation: training (Llama vs. R1-Distill) changes both encoding and behavior; inference-time thinking (Qwen no-think vs. think) changes behavior without changing encoding.

Selectivity. Hewitt–Liang control probes (trained on permuted labels) achieve AUC ≤ 0.66 at all layers for both models, confirming that the true probes decode representational content, not probe capacity. Selectivity exceeds 34 percentage points at peak layers for both models.[1]

Immune category control. Probes trained on the four immune categories (CRT, arithmetic, framing, anchoring) achieve AUC = 1.0 at layers 0–1 for both models, declining to AUC ≈ 0.95 at deeper layers. This *complicates* the interpretation: the probes partially detect the presence of lure text (i.e., the conflict/control distinction in the stimulus itself) rather than exclusively reflecting the model's processing mode. We address this confound in Section **??**.

4.3 Summary of Probing Results

Table **??** summarizes the key probing metrics.

4.4 Resolving the Surface-Feature Confound

Probes achieve AUC = 1.0 on behaviorally immune categories (CRT, arithmetic, framing, anchoring) at early layers, raising the concern that the conflict/control signal partially reflects surface features of lure text rather than processing mode. We resolve this confound with three converging analyses.

Cross-prediction. We train a probe on Llama's *vulnerable* categories (base rate, conjunction, syllogism) and test it on the *immune* categories (CRT, arithmetic, framing, anchoring). If the probe captures a universal processing-mode signal, it should transfer; if it captures category-specific lure susceptibility, it should not. At the peak layer (L14), the Llama probe achieves transfer AUC = 0.378—below chance parity and far below the

[1] We report two complementary probe analyses: (i) 5-fold CV on all 330 items across 7 categories, which peaks at L14 with AUC = 0.999 (Llama) and is used for cross-prediction; (ii) bootstrap CI analysis restricted to 160 vulnerable-category items (3 categories, 1000 resamples), which peaks at L16 with AUC = 0.974 [0.952, 0.992]. The bootstrap analysis provides statistical rigor; the all-category analysis provides the direction used for cross-prediction and steering.

within-category AUC of 0.987. The probe is specific to the categories where the model is actually vulnerable. For R1-Distill, results are mixed: transfer AUC = 0.878 at L4 (early, likely surface features) but 0.385 at L31 (late, consistent with specificity).

Transfer matrix. Training probes on each vulnerable category and testing on all others (Llama, L14) reveals a structured pattern: base rate and conjunction probes transfer near-perfectly to each other (AUC > 0.99), while transfer to syllogism is weaker (0.59–0.63), and CRT probes transfer poorly to vulnerable categories (AUC = 0.06–0.45). This suggests a shared processing-mode direction for base rate and conjunction that is partially shared with syllogism but absent in immune categories.

Lure susceptibility scores. We extract continuous `P0` logit scores for the S1-lure answer on each conflict item. Llama's mean lure susceptibility is +0.422 (the lure answer is favored in the pre-generation representation), while R1-Distill's mean is −0.326 (the correct answer is favored). The sign flip confirms that the representational difference is not just about separability of conflict/control conditions but about the *direction* of the model's initial commitment: the standard model leans toward the lure before generation begins; the reasoning model leans away from it.

Cross-model probe transfer. Cross-model transfer further validates the signal: a probe trained on Llama's vulnerable categories achieves AUC = 0.920 when tested on R1-Distill's activations at layer 23 (vs. within-model AUC of 0.963). The reverse transfer—R1-Distill to Llama—achieves AUC = 0.954 at layer 15 (vs. self AUC of 0.919). The S1-like/S2-like processing-mode direction is substantially shared across models that differ in reasoning training but share architecture and vocabulary, contrasting with **?**'s near-orthogonal finding on architecturally distinct models.

4.5 SAE Feature Analysis

Sparse autoencoder analysis of Llama's layer 19 residual stream (Goodfire SAE, 65,536 features) identified 41 features with statistically significant differential activation between conflict and control items (BH-FDR $q < 0.05$). All 41 features survived Ma et al. falsification testing (**?**): injecting top-activating tokens into random text did not reproduce the activation pattern. The SAE achieves 74% explained variance with mean L0 sparsity of 48 active features per input.

4.6 Attention Head Specialization

Per-head Mann-Whitney U tests (BH-FDR $q < 0.05$) on attention entropy reveal that R1-Distill has nearly twice as many S2-like-specialized heads—heads with significantly higher entropy on conflict items and effect size $|r_{rb}| \geq 0.3$—as Llama (57 vs. 30 out of 1024 total heads, 5.6% vs. 2.9%). Reasoning training appears to increase the proportion of attention heads that behave differently on deliberation-requiring items.

5 Discussion

Conflict detection without resolution. The central finding maps precisely onto De Neys' conflict detection paradigm (**??**): the standard model *detects* the conflict between heuristic and normative responses (AUC = 0.974 at L16) yet *fails to resolve* it (27.3% lure rate). This parallels the human finding that conflict monitoring—indexed by anterior cingulate cortex activation (**??**)—often occurs even when the heuristic response prevails. Our `P0` probe functions analogously to an ACC conflict signal: it registers the presence of a heuristic–normative tension in the pre-generation representation, but this tension alone is insufficient to override the heuristic pathway. The first-token confidence data sharpen this parallel: Llama's reduced first-token probability (0.751 vs. 0.793) and elevated entropy (0.948 vs. 0.817) on conflict items provide a behavioral analogue of De Neys' response-time signatures, confirming that the model's output reflects the internal tension even before chain-of-thought unfolds.

The reasoning model shows *less* distinct conflict encoding (AUC = 0.930) yet dramatically better resolution (2.4% lure rate)—a pattern confirmed both behaviorally and mechanistically in the OLMo family (14.9% →

0.9% lure rate; AUC 0.996 → 0.962, non-overlapping 95% CIs, higher in 30/32 layers), establishing this as a general phenomenon across architectures. If reasoning training simply added a "System 2 module" that overrides System 1, we would expect sharper separability. Instead, the AUC drop suggests that reasoning distillation makes Type 2 processing *autonomous*—a phenomenon Evans explicitly predicts: "Type 2 processes, practiced over time, can become automatic and default" (**??**). On this view, R1-Distill does not override S1 with S2; it has internalized the deliberative pathway so thoroughly that it no longer needs a sharp conflict signal to route items correctly—consistent with evidence that base models already possess latent reasoning capabilities, with post-training teaching *when* to deploy them rather than *how* (**?**). The lure susceptibility scores confirm this: Llama's pre-generation representation actively favors the lure (+0.422), while R1-Distill's favors the correct answer (−0.326). Reasoning training doesn't just blur a boundary—it flips the model's default commitment.

Domain-specific vulnerability. Three categories are vulnerable (base rate neglect, conjunction fallacy, syllogisms) while four are immune (CRT, arithmetic, framing, anchoring). This category-specificity mirrors Stanovich's concept of *dysrationalia* (**?**): rational thinking deficits are domain-specific, not general. Just as high-IQ humans can fail base rate problems while acing CRT variants, our models show selective vulnerability that is specific to certain heuristic families. The OLMo replication sharpens this point: OLMo Instruct is vulnerable to loss aversion (33.3%) where Llama is immune (0%), yet both share base rate and conjunction vulnerability. Bias profiles are model-specific—a function of training data and optimization procedure, not architecture—which means auditing a single model family is insufficient for characterizing LLM bias susceptibility in general.

Ecological rationality and format dependence. A preliminary test of Gigerenzer's ecological rationality critique (**?**) yielded a striking reversal: natural frequency framing *increased* lure susceptibility in both models (Llama: 84% → 100%; R1-Distill: 4% → 40%), suggesting that LLM bias sensitivity is format-dependent in ways that diverge from human cognition, where natural frequencies reliably *reduce* base rate neglect. This inversion warrants systematic investigation across all bias categories and models.

Training vs. inference. Qwen's think/no-think result is the sharpest finding: thinking mode reduces lure rates from 21% to 7.3% on the same weights, yet probes are identical in both modes (AUC = 0.971 at L34). The toggle changes what the model *does* during generation, not what it *encodes* at `P0`. Combined with the Llama/R1-Distill comparison (where training *does* change encoding: 0.974 → 0.930), this yields a clean dissociation: **training changes S1-like/S2-like encoding; inference-time thinking changes S1-like/S2-like behavior without changing encoding.**

This parallels the neuroscience distinction between structural and functional plasticity (**?**): training rewires the weights so that P0 favors the normative answer; inference-time thinking adapts the activation pattern on an unchanged structural substrate. This is consistent with **?** and **?**'s overlapping subspaces, but adds a new dimension: the encoding/behavior dissociation reveals two distinct mechanisms for bias resistance, not one.

Specificity. Cross-prediction (Section **??**) resolves the surface-feature confound: Llama's vulnerable-category probe yields transfer AUC = 0.378 on immune categories, and the transfer matrix shows base rate and conjunction share a processing-mode direction (mutual AUC > 0.99) partially shared with syllogism (0.59–0.63) but absent in CRT (AUC = 0.06–0.45). Cross-model transfer (Llama→R1: 0.920; R1→Llama: 0.954) confirms that this direction is shared across reasoning-differentiated models, not model-specific. The probe captures how the model processes lures, not that lures are present.

Relation to CogBias. **?** showed that bias-related directions are *steerable* (26–32% bias reduction via activation steering); we show the same directions are *reorganized* by reasoning training. This suggests a practical strategy: steer standard models along the directions that reasoning training blurs, approximating the reasoning model's robustness without chain-of-thought cost. We extend CogBias methodologically via (i) continuous lure susceptibility scores, (ii) cross-category transfer analysis, (iii) a within-model reasoning toggle (Qwen think/no-think), and (iv) cross-model probe transfer showing the S1-like/S2-like direction

is shared when architecture is held constant (AUC > 0.92 both directions), in contrast to CogBias's near-orthogonal finding across architecturally distinct models.

Generation strategy dependence. Multi-seed evaluation with sampled decoding (temperature 0.7) reveals that behavioral vulnerability profiles are generation-strategy-dependent. Under sampling, Llama's overall lure rate remains stable (27.5% $\pm$ 1.3pp across 3 seeds), but the category profile shifts: framing rises from 0% (greedy) to 53%, CRT from 0% to 36%, while conjunction drops from 55% to 3%. This suggests that the "immune" categories are immune only under greedy decoding, and that our primary results should be interpreted as reflecting the model's maximum-probability behavior rather than its distributional behavior. Crucially, the probe results at `P0` are unaffected by generation strategy, as they measure the pre-decision representation before any tokens are generated.

Scale effects. At 32B scale (OLMo-3.1), vulnerability *increases* (19.6% vs. 14.9% at 7B) while reasoning training remains effective (0.4% Think vs. 0.9% at 7B). Probe separability at 32B is near-perfect (AUC $= 0.9999$ at L20/64), confirming these signatures are not small-model artifacts. OLMo-32B-Think probes reach AUC $= 0.9978$ ($\Delta\, 0.0021$ from Instruct), suggesting vulnerability encoding persists at larger scale.

Methodological safeguards. Probing can produce plausible results even on random networks (**?**), and SAE features often fail falsification (**?**). We address this five ways: (1) Hewitt–Liang controls (Section **??**) show selectivity >34pp; (2) cross-prediction (Section **??**) provides structural falsification—noise probes would not systematically drop to AUC $= 0.378$ on immune categories; (3) the inter-model comparison is itself a control—identical prompts yield different AUCs (0.974 vs. 0.930) only if the models encode inputs differently; (4) cross-model probe transfer (AUC > 0.92) confirms the signal is not model-specific but reflects a shared representational direction; (5) all 41 SAE features survive Ma et al. falsification (Section **??**), with zero spurious features after token-injection testing.

Limitations. (i) Model scale: results extend to 32B (OLMo-3.1) but not frontier-scale models (>100B); the blurring effect may manifest differently at that scale. (ii) Linear probes may miss nonlinear representational structure; MLP probes and CCS (**?**) are planned robustness checks. (iii) We measure conflict/control separability, which is a proxy for S1-like/S2-like processing mode—not a direct readout of "how the model is thinking." (iv) The Qwen think-mode `P0` position is pre-think; probing *within* the thinking trace (T0 through Tend) would reveal whether representations shift during chain-of-thought.

Future work. SAE analysis has identified 41 falsification-surviving features that differentiate conflict and control processing (Section **??**). The natural next step is **causal interventions**: steer these features in the standard model to test whether artificially blurring the S1-like/S2-like boundary improves bias resistance without chain-of-thought cost, and conversely, whether sharpening the boundary in the reasoning model reintroduces vulnerability.

Acknowledgments

[Redacted for anonymous review.]

Code and Data. Code, benchmark, and activation data will be released at [URL upon acceptance].